

Equation Sample 5

Dataset-expansion sample

1 Derivation

The following display is drawn from a source-backed image-to-Latex benchmark and reproduced verbatim.

$$\begin{aligned}v_{mn}^x &= \langle m|v^x|n\rangle \\&= \delta_{mn}\langle v^x\rangle\left(\frac{k_x}{\rho}\right)(-\delta_{m,0}+\delta_{m,1})+(1-\delta_{mn})\langle v^x\rangle\left(\frac{k_z}{\rho}\frac{k_x}{\sqrt{k_x^2+k_y^2}}-i\frac{k_y}{\sqrt{k_x^2+k_y^2}}\right) \\&= \delta_{mn}\langle v^x\rangle(\sin\theta\cos\phi)(-\delta_{m,0}+\delta_{m,1})+(1-\delta_{mn})\langle v^x\rangle(\cos\theta\cos\phi-i\sin\phi), \\v_{mn}^y &= \delta_{mn}\langle v^y\rangle(\sin\theta\sin\phi)(-\delta_{m,0}+\delta_{m,1})+(1-\delta_{mn})\langle v^y\rangle(\cos\theta\sin\phi-i\cos\phi), \\v_{mn}^z &= \delta_{mn}\langle v^z\rangle\cos\theta(-\delta_{m,0}+\delta_{m,1})+(1-\delta_{mn})\langle v^z\rangle\frac{-\sin\theta}{\rho},\end{aligned}$$